CSE315:Introduction to Data Science

WEEK-3

Statistics Introduction

- Statistics is the study of the collection, organization, analysis, interpretation, and presentation of data. No statistics are possible without data.

Types of Statistics

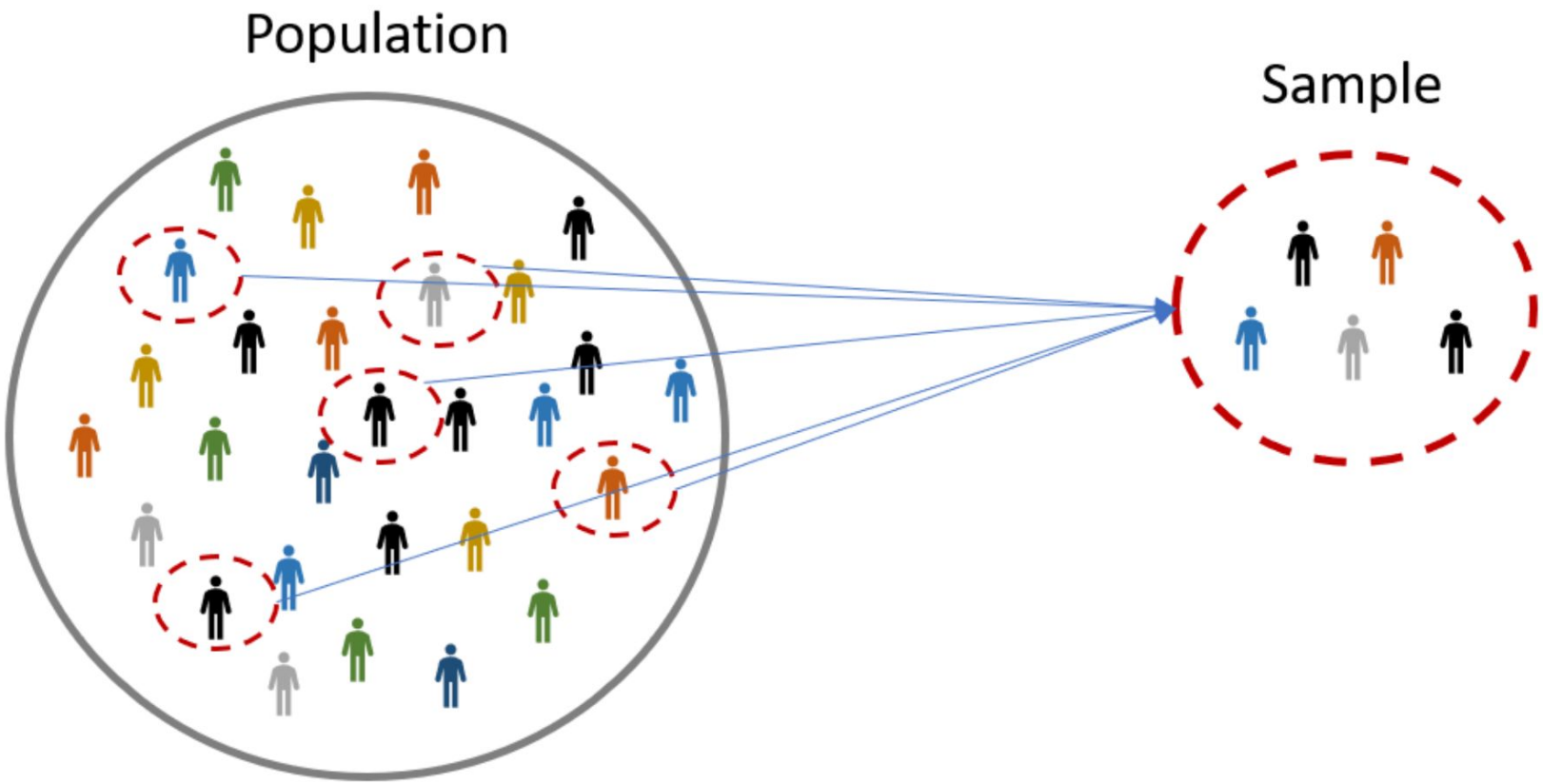
- **Descriptive Statistics:** The data is described by descriptive statistics.
- **Inferential Statistics:** Inferential statistics help in decision making by examining (Test of hypothesis) the project.

Types of Data used in statistics

- The data used in statistics can be of two types
- **Primary Data** - When a statistician collects data by himself or through his own team it is called primary data.
- **Secondary Data** - Secondary data are data collected from any other source. Secondary data is not collected by the statistician himself, but from any other statistic or source. Secondary data source is always to be mentioned.

Statistical Method

Population and Sample



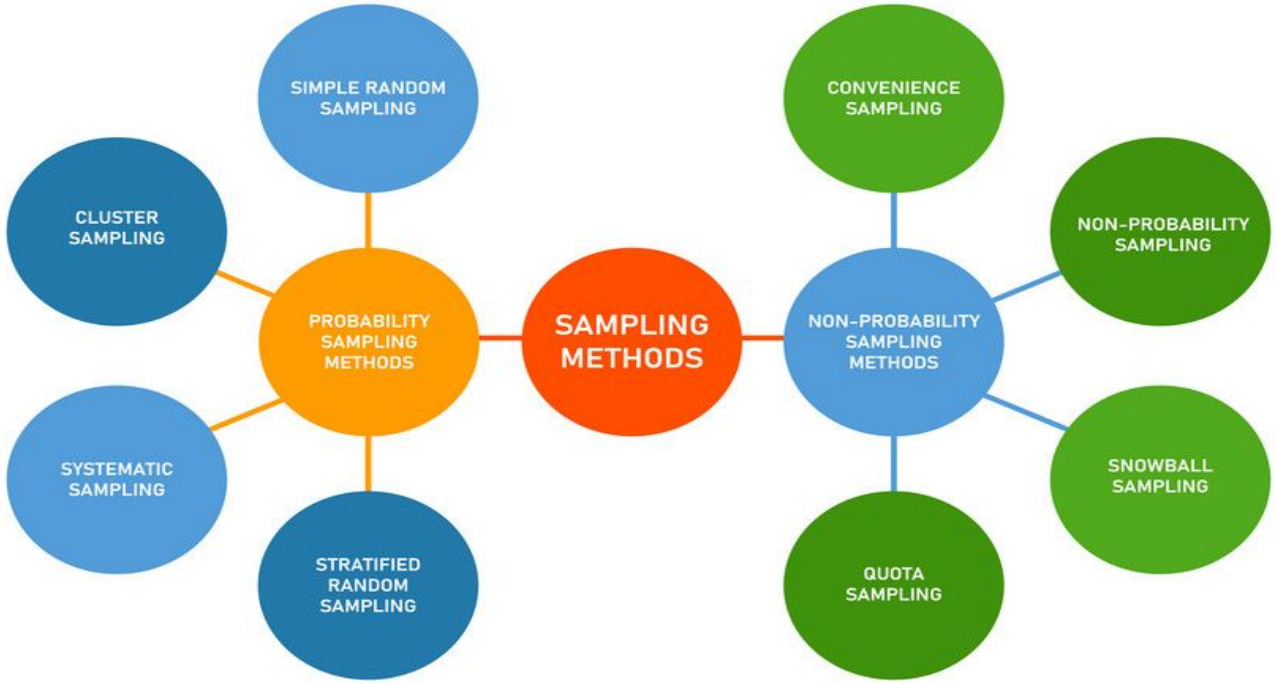
Sample from Population

- The sample must be representative, that is, the sample must represent the whole population.
- **Example**: If an ordinary person is asked to pick a mango from a basket, he will only choose a big, fresh and ripe one. As a result, the collected specimens became one-sided, meaning that not all of these specimen represent different types of mango. So the sample has to be selected in such a way that there is no chance of bias.
- If we **do not select the correct sample** then the results of the study will be **wrong**, so sampling is considered important in any study.

Necessity of Sampling

- Suppose you want to know the average salary of all the IT professionals in Bangladesh, that is, all the IT professionals in Bangladesh are the **population**.
- In that case you have to go to all the IT professionals in Bangladesh and find out their salary and the average that will be obtained from the data obtained will be the actual average.
- When the data of all the members of the population is taken it is called **census survey**.
- In fact, in most cases it is not possible to collect data from all members of the population. **Because it requires a lot of time, manpower and money.** That is why a **small part** representing the population is taken instead of the whole population.

Types of Sampling



Probability and Non-probability Sampling

- **Probability Sampling** - Probability Sampling has the probability of being selected as a sample of each member of the population.
- **Non-Probability Sampling** - This method is called non-probability sampling because the researcher's judgment is important in selecting samples in non-probability sampling. Non-probability sampling usually does not use probability mechanism. The decision of the researcher is the rule in this case.

Types of Probability Sampling

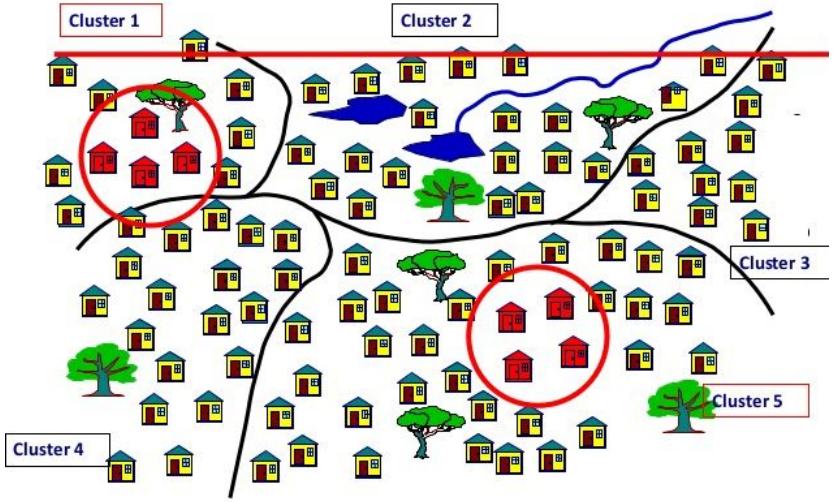
- Simple Random Sampling
- Cluster Sampling

Simple Random Sampling

- **Simple Random Sampling** - The most fundamental method of probability sampling is simple random sampling. In this method a certain number of samples are selected from the population through probability mechanism.
- **For example** lottery ticket. Each ticket in the basket is likely to be selected at the time of ticket selection. Simple random sampling can also be done through random number generation.

Cluster Sampling

- Cluster sampling- Cluster sampling is called geographical location based sampling. For example, if a district-upazila union based sample is taken in Bangladesh, it will be a cluster sample.
- Cluster samples can also be taken from other geographical divisions. Cluster sampling can again be divided into two parts, one stage cluster and two stage cluster.



One and two stage cluster sampling

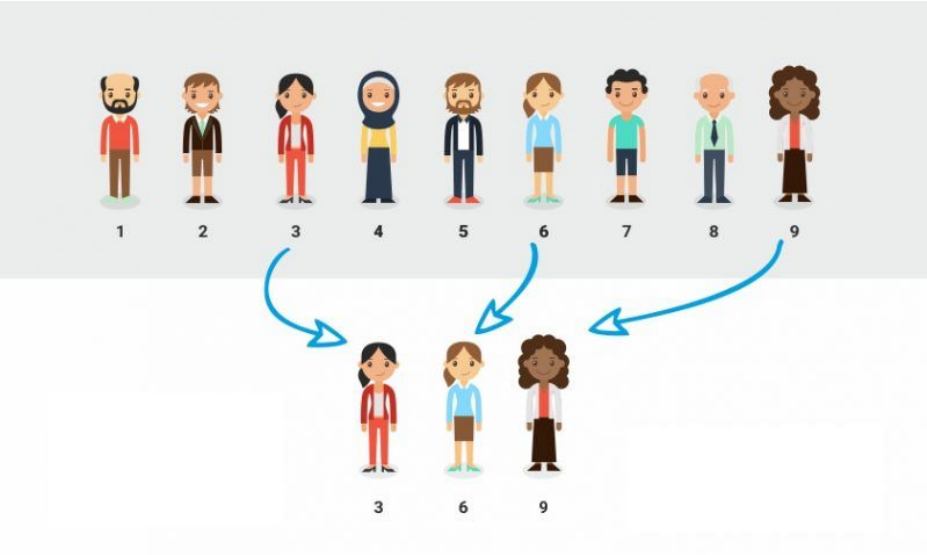
- When all the components inside a cluster are taken as samples it is a one stage cluster. Suppose each ward of Dhaka city is considered as a cluster, in which case samples of all the people of the ward will be taken in one stage cluster.
- In the case of two stage clustering, instead of taking everyone inside a cluster as a sample, a certain number of samples are selected by following a simple random sampling method. In other words, we want to take data of 100 people from each cluster so 100 people will be selected again from each cluster through simple random sampling.

Stratified Sampling

- **Stratified Sampling** - Stratified sampling is a homogenous group based sampling method. All members of the same group usually have the same characteristics, such groups are called strata.
- **For example**, we want to **find out the average age** of students in a class. This is why we can **create two separate groups** of female students and male students, so that it is possible to **take separate samples from each homogenous group** and each sample represents a group whose members are all homogenous or possess the same characteristics.

Systematic Sampling

- **Systematic Sampling** - Systematic sampling follows a specific sequence. For example, every 2, 3, 4 or any number of data is taken. For example, if a researcher wants to study the number of family members living in a ward of Dhaka city, he wants to collect data of 100 families through systematic sampling.

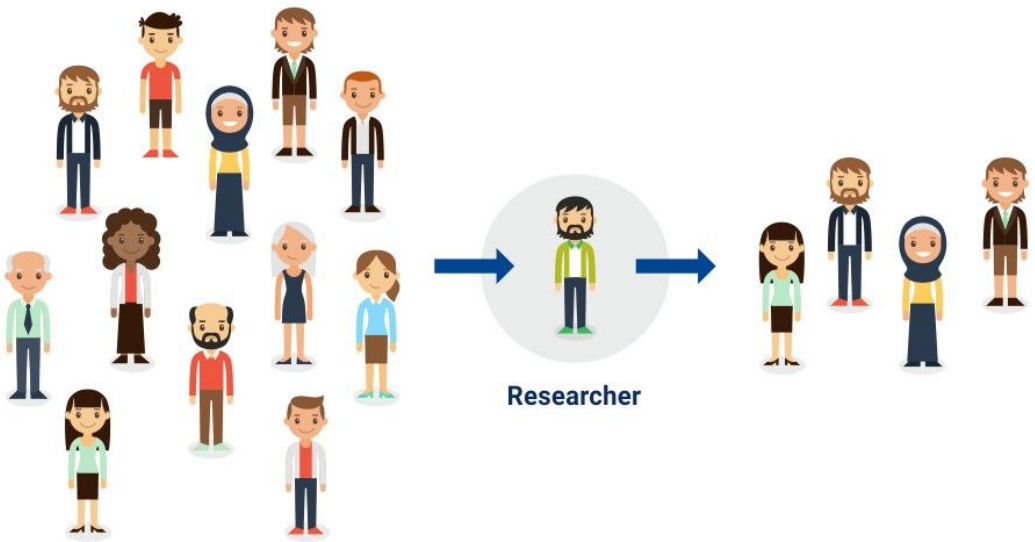


Types of Non-probability Sampling

- Judgmental Sampling
- Snow-Ball Sampling
- Convenience Sampling
- Quota Sampling

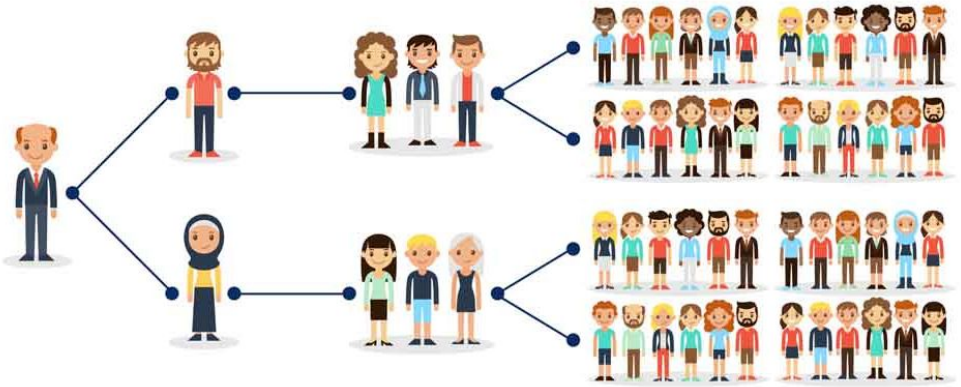
Judgmental Sampling

- **Judgmental Sampling** - In judgmental sampling, the researcher selects samples from the population according to his own discretion.



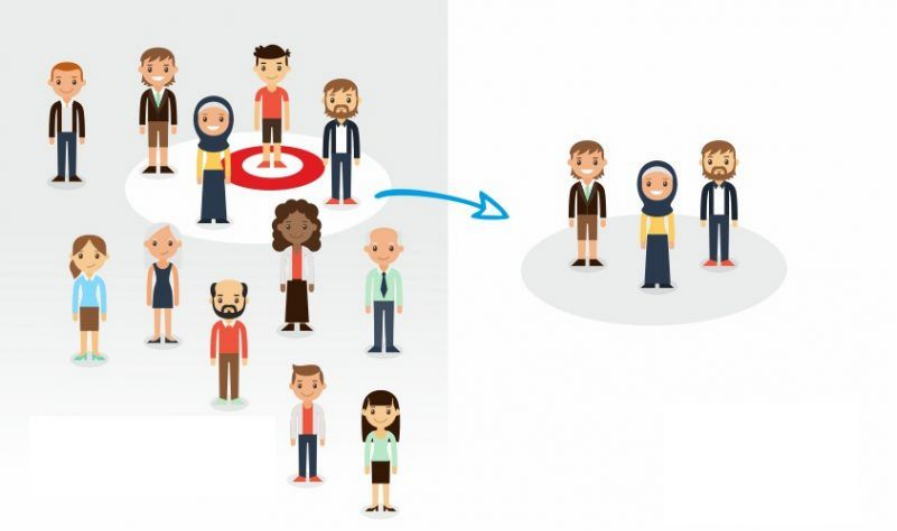
Snow-Ball Sampling

- **Snow-Ball Sampling** - It usually takes samples that are difficult to reach in parts of the population. In this case a sample serves as a reference to the next sample.
- **For example**, it is naturally difficult to find HIV patients or drug addicts because in most cases such people hide their identities or find them difficult to find. In such cases samples are collected by means of snow-ball sampling, in which case the reference is passed from one person to another, thus gradually reaching the entire population.



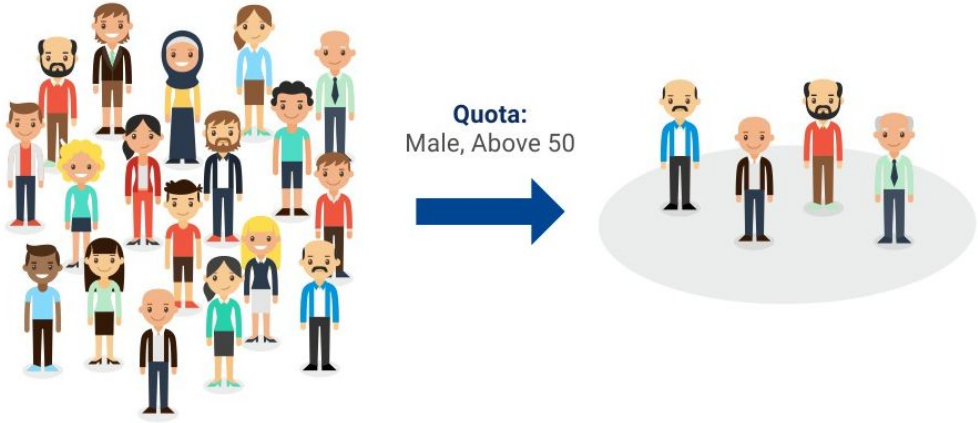
Convenience Sampling

- **Convenience Sampling** – It allows the researcher to select a member from his or her hand or a relatively readily available sample.
- **For example**, if a company wants to give feedback to customers about a product that is newly marketed, then feedback is taken from the buyers who are available in the market.



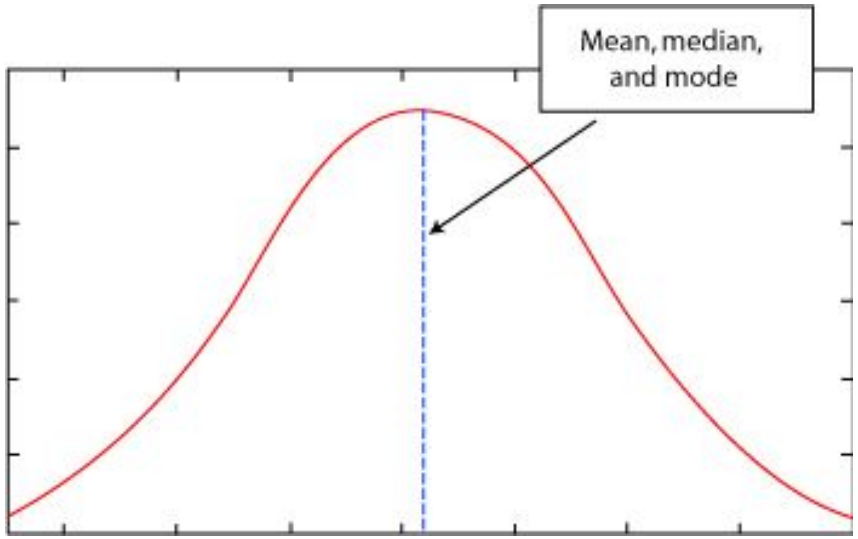
Quota Sampling

- **Quota Sampling** - In the case of quota sampling, usually a certain proportion or quota quota is followed in the sample selection.
- **For example**, if 80% of the population is male and 40% female, then the same ratio will be followed during sample selection.



Central Tendency

- Mean, mode and median is a measure of the central tendency of any data.
- Many of us look at the shape of a mountain peak, sand dune or bell shape when we format any data. Most of the data is concentrated in the middle. This tendency to have data in the middle is called central tendency.



Mean

- **Mean** - Mean is average. In simple words, the sum of all the numbers is divided by the total number of numbers.
- **Suppose** the age of 10 students in a class is 14,13,11,14,15,12,16,17,11,14 respectively.

Data

14	13	11	14	15	12	16	17	11	14

$$\text{Mean } \bar{x} = \frac{\sum x}{n} = \frac{14+13+11+14+15+12+16+17+11+14}{10} = 13.7$$

That means we can say that the average age of students in that class is 13.7 years.

Mode

- **Mode-** Mode is the most common number or the number that comes up the most times.

Data

14	13	11	14	15	12	16	17	11	14

Data	Frequency
14	3
13	1
11	2
15	1
12	1
16	1
17	1

Number 14 comes up most (3 times).
So, Mode=14

Median

- **Median** - Median is called the middle value. Median is the middle value that is obtained by sorting the data. The median value is easily obtained if the total observation of the data is an odd number, but if the observation is an even number then the average of any two data in the middle is taken as median.
- If the no. of observations are odd, then $(\frac{n}{2}+1)th$= 5th data= 14
- If the no. of observations are even, then $\frac{\frac{n}{2}th+\left(\frac{n}{2}+1\right)th}{2} = \frac{14+14}{2} = 14$

Data

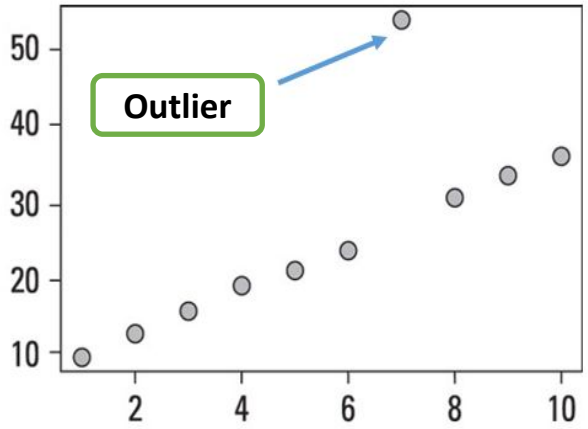
14	13	11	14	15	12	16	17	11

Sorted Data

11	11	12	13	14	14	15	16	17

Why need median when mean has

- The question that may come to your mind is why do you need median again to have mean?
- The median is not affected by outlier.
- **Outlier** - outlier is outsourced data, which means it is not like others. Whose value is much lower or much higher than most values.
- Outliers may change the mean, that is, average values may decrease or increase, but **outliers do not affect the median value**. That is why the media is needed.



Data Spread

- **Range** - Range is the sum of the maximum and minimum values of the data.
- For example, the maximum value is 18 and the minimum value is 11 and the difference between them is 8, this is the range. This shows how much data is stored.
- Range= Maximum-Minimum= 17-11=6

Data

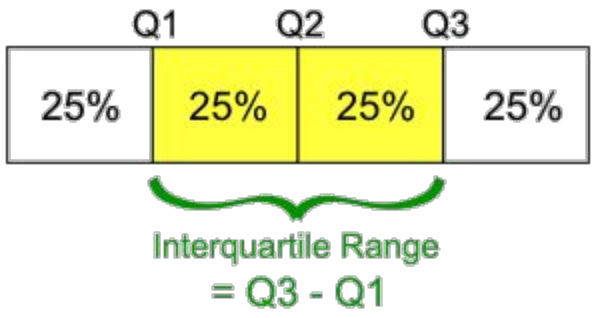
14	13	11	14	15	12	16	17	11

Sorted Data

11	11	12	13	14	14	15	16	17

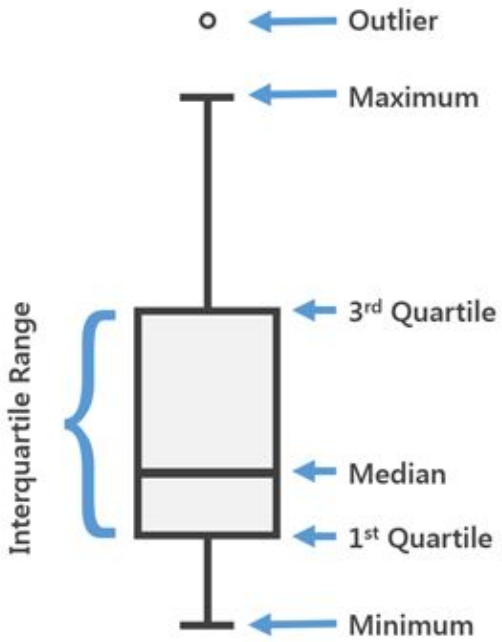
Data Spread

- **Quartiles and IQR** - Data is divided into **4 parts** by quartiles. Each section contains 25% data. It contains 50% data in the second quartiles and 25% data in the first and last quartiles.
- The subtraction of 3rd quartile and 1st quartile is called inter quartile range or IQR.



Box Plot

- If a data box is plotted, maximum and minimum value of data, outlier, IQR etc. can be visualized.



Finding the Outliers of a Data Set

- **Outlier:** An outlier is an observation that lies an abnormal distance from other values in a random sample from a population. In a sense, this definition leaves it up to the analyst (or a consensus process) to decide what will be considered abnormal. Before abnormal observations can be singled out, it is necessary to characterize normal observations.

Finding the Outliers of a Data Set

- The table shows the heights in meters, of the tallest buildings in a city. If there are any outliers in the data, find their values.

607	630	762	685	714	561
678	662	550	901	502	725

- we can use the 1.5-times-IQR rule. An outlier x would be less than $Q1$ minus 1.5 times the IQR or, or the outlier would be greater than $Q3$ plus 1.5 times IQR.

$$1.5 \times IQR\ Rule:$$

$$x < Q1 - 1.5 \times IQR$$

$$x > Q3 + 1.5 \times IQR$$

Finding the Outliers of a Data Set

- Our first step here is to calculate the interquartile range and find these boundaries. And to do that, the first thing we do is put the data in size order.

502 550 561 607 630 662 678 685 714 725 762 901

- We also know that each quartile is 25 percent of the data. Quartile: 25%. Since we have 12 building heights, we can divide 12 by four, which is three. And that means our first quartile will occur after the third data point, our second quartile after the sixth data point, and our third quartile after the ninth data point.

502	550	561	607	630	662	678	685	714	725	762	901

Finding the Outliers of a Data Set

502	550	561	607	630	662	678	685	714	725	762	901

Q1=584 Q2 Q3=719.5

$$=\frac{561+607}{2} \qquad =\frac{714+725}{2}$$

IQR=Q3-Q1

= 719.5-584

= 135.5

So, Q1 = 584, Q2 = 719.5, IQR = 135.5

Finding the Outliers of a Data Set

- We're now ready to go back and use these rules to calculate the upper and lower bounds for outliers.
- Lower Bound for Outliers:

$$x < Q1 - 1.5 \times IQR$$
$$x < 584 - 1.5 \times 135.5$$
$$x < 380.75$$

- Upper Bound for Outliers:

$$x > Q3 + 1.5 \times IQR$$
$$x > 719.5 + 1.5 \times 135.5$$
$$x > 922.75$$

In order for there to be an outlier on the upper end, it would need to be greater than 922.75. Our largest data point is 901, which is less than this value. And since none of our data values are less than the lower bound for the outliers or greater than the upper bound for outliers, there are no outliers in this data set.

Finding the Outliers of a Data Set

- **Exercise:**

Find all possible outliers for the following set of data:

108, 31, 75, 87, 79, 88, 89, 118, 51, 89, 174, 95, 51, 70, and 73.

Exercise:

Your dataset has 11 values. You have a couple of extreme values in your dataset, so you'll use the IQR method to check whether they are outliers.

26 37 24 28 35 22 31 53 41 64 29